

LABORATORY OBSERVATION

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 3

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1. TITLE

Implementation of Basic Student Profile using JavaScript Classes and DOM

2. AIM

To construct an interactive student profile generator using JavaScript ES6 classes to encapsulate student data, and DOM manipulation techniques to dynamically render the data on the webpage.

3. OBJECTIVE

The objectives of this experiment are:

- To define JavaScript ES6 classes with constructors for structuring data.
 - To instantiate objects from defined classes using user-provided input.
 - To manipulate the Document Object Model (DOM) to capture user input values.
 - To dynamically inject HTML content and display structured object properties on the webpage.
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4. THEORY

JavaScript ES6 introduced the `class` keyword, providing a much simpler and clearer syntax to create objects and deal with inheritance. A class acts as a blueprint for creating objects, containing a `constructor` method to initialize properties.

The Document Object Model (DOM) connects web pages to scripts. Methods like `getElementById` allow developers to fetch user input, while the `innerHTML` property allows dynamic insertion of HTML nodes into the document to display output.

5. ALGORITHM / PROCEDURE

1. Create an HTML layout with input fields for Name, Roll No, Department, and CGPA.
 2. Add a styled button configured to trigger a JavaScript function `showProfile()` on click.
 3. Define an ES6 class `Student` with a constructor to initialize properties.
 4. In the function, retrieve the values from the input fields using DOM methods.
 5. Implement validation to alert the user if any fields are empty.
 6. Instantiate a new `Student` object passing the input values.
 7. Use template literals to construct HTML strings mapping the object's properties.
 8. Update the `innerHTML` of the profile display container to show the student details.
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6. PROGRAM

Source File: task3.html

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Basic Student Profile</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      padding: 20px;
      background: #f5f5f5;
    }
    .container {
      max-width: 400px;
      margin: auto;
      background: white;
      padding: 25px;
      border-radius: 10px;
      box-shadow: 0 2px 10px rgba(0,0,0,0.1);
    }
    input {
      width: 100%;
      padding: 8px;
      margin: 5px 0 15px 0;
      border: 1px solid #ddd;
      border-radius: 5px;
      box-sizing: border-box;
    }
    button {
      width: 100%;
      padding: 10px;
      background: #007bff;
      color: white;
      border: none;
      border-radius: 5px;
      cursor: pointer;
      font-size: 16px;
    }
    button:hover {
      background: #0056b3;
    }
    .profile {
      margin-top: 20px;
      padding: 15px;
      background: #f9f9f9;
      border-radius: 5px;
      border-left: 4px solid #007bff;
    }
    .profile p {
      margin: 8px 0;
    }
    .label {
      font-weight: bold;

```

```

        display: inline-block;
        width: 100px; /* Keeps the colons aligned */
    }
</style>
</head>
<body>

<div class="container">
    <h2>Student Profile</h2>

    <label>Name</label>
    <input type="text" id="name" value="A.Rohit">

    <label>Roll Number</label>
    <input type="text" id="roll" value="A23126552001">

    <label>Department</label>
    <input type="text" id="dept" value="CSE (AI & ML)">

    <label>CGPA</label>
    <input type="text" id="cgpa" value="8.42">

    <button onclick="showProfile()">Show Profile</button>

    <div id="profileDisplay"></div>
</div>

<script>
    // 1. Create the Student class with properties
    class Student {
        constructor(name, rollNumber, department, cgpa) {
            this.name = name;
            this.rollNumber = rollNumber;
            this.department = department;
            this.cgpa = cgpa;
        }
    }

    // 2. The function that runs when the button is clicked
    function showProfile() {

        // Get user input values from the text boxes
        let name = document.getElementById('name').value;
        let roll = document.getElementById('roll').value;
        let dept = document.getElementById('dept').value;
        let cgpa = document.getElementById('cgpa').value;

        // Validation - stop if fields are empty
        if (name === '' || roll === '' || dept === '' || cgpa === '') {
            alert('Please fill in all fields.');
```

```
        return;
```

```
    }
```

```
    // 3. Create a new Student object
```

```
let student = new Student(name, roll, dept, cgpa);

// 4. Display profile dynamically using simple innerHTML
let display = document.getElementById('profileDisplay');
display.innerHTML = `
    <div class="profile">
        <p><span class="label">Name</span> : ${student.name}</p>
        <p><span class="label">Roll No</span> : ${student.rollNumber}</p>
        <p><span class="label">Department</span> : ${student.department}</p>
        <p><span class="label">CGPA</span> : ${student.cgpa}</p>
    </div>
`;
    }
</script>

</body>
</html>
```

7. CODE EXPLANATION

Code / Syntax	Explanation
class Student {}	Defines a JavaScript ES6 class acting as an object blueprint.
constructor(...)	Special method for creating and initializing an object created with a class.
document.getElementById()	DOM method that returns the element with the specified ID.
element.value	Property that sets or returns the value of an input field.
new Student(...)	Instantiates a new object instance of the Student class.
element.innerHTML	Sets or returns the HTML content (inner markup) of an element.

8. TEST CASES AND EXECUTION DATA

The experiment was executed with the following test cases to verify functionality:

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Empty Input	Click button with empty fields	Triggers alert warning for empty fields	Pass

TC-02	Object Init	Enter details & submit	Instantiates Student class successfully	Pass
TC-03	DOM Injection	Verify profile output section	HTML dynamically displays entered values	Pass

9. OUTPUT

Output 1

Student Profile

Name
A.Rohit

Roll Number
A23126552001

Department
CSE (AI & ML)

CGPA
8.25

Show Profile

Name : A.Rohit

Roll No : A23126552001

Department : CSE (AI & ML)

CGPA : 8.25

10. RESULT

Thus, the basic student profile generator utilizing JavaScript ES6 classes and DOM manipulation was successfully implemented and verified.